BTS -VI(SS)-11.14-1115 Reg. No. [ ]

A

B. Tech. Degree VI Semester Special Supplementary Examination November 2014

ME 602 DYNAMICS OF MACHINERY

(2006 Scheme)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

I. (a) State and explain D'Alembert's principle in the analysis of dynamics of machinery.
(b) Explain how the friction at joints of a mechanism is handled in force analysis.
(c) Explain coefficient of fluctuation of speed and coefficient of fluctuation of energy.
(d) Explain gyroscopic effect on ships.
(e) Explain condition of complete balancing of an engine.
(f) Explain hammer blow and variation in tractive effort.
(g) Explain centrifugal tension and initial tension in belt drives.
(h) Explain band and internal expanding brakes.

PART B

(4 x 15 = 60)

II. A four-link mechanism is subjected to following external forces. Determine the shaft torque T on the input link AB for static equilibrium of the mechanism. Also find the forces on the bearings A, B, C and D.

Link	Length (mm)	Force (N)	Point of application
AB	500	$80 < 73.5^{\circ}$	325 mm from A
BC	600	$144 < 58^{\circ}$	297 mm from B
CD	560	$60 < 42^{\circ}$	373 mm from D
AD	1000	Fixed link	

OR

III. The dimensions of a four link mechanism are AB = 500mm, BC = 660 mm, CD = 560 mm and AD = 1000 mm. The link AB has an angular velocity of 10.5 rad/sec counter clockwise and an angular retardation of 26 rad/s^2 at the instant when it makes an angle of 60° with AD, the fixed link. The mass of the links BC and CD is 4.2 kg/m length. The link AB has a mass of 3.54 kg, the centre of which lies at 200 mm from A and a moment of inertia of 88500 kg.mm^2. Neglecting gravity and friction effects, determine the instantaneous value of the drive torque required to be applied on AB to overcome the inertia forces.

(P.T.O.)

IV. The torque exerted on the crank shaft of a two stroke engine is given by T(N.m)=14500+2300 $\sin^2\beta - 1900\cos^2\beta$ where β is the crank angle displacement from the inner dead centre. Assuming the resisting torque to be constant, determine: (i) the power developed when the speed is 150 rpm (ii) the moment of inertia of the fly wheel if the total fluctuation of speed is not to exceed 1% of the mean speed; (iii) the angular acceleration of fly wheel when the crank has turned through 30° from the inner dead centre.

OR

V. A car is of total mass 2000 kg. It has wheel base equal to 2.4 m and track width = 1.4 m. The centre of gravity lies at 500 mm above ground level on the vertical line through geometric centre of the wheels. The effective diameter of each wheel is 800 mm and mass moment of inertia of each wheel is 1 kgm^2. The rear axle ratio is 4. The mass moment of inertia of engine rotating parts is 3 kgm^2. The spin axis of the rotating engine parts is parallel to the spin axis of the wheels. The engine is rotating in the same sense as the wheels. Determine the critical speed of the car when it takes a right turn of 100 m radius.

VI. Four masses are attached to a shaft at planes A, B, C and D at equal radii. The distance of the planes B, C and D are 40 cm, 50 cm and 120 cm respectively. The masses at A, B and C are 60 kg, 45 kg and 70 kg respectively. If the system is in complete balance, determine the mass at D and the position of the masses B, C and D with respect to A.

OR

VII. The cranks of a two-cylinder, uncoupled inside cylinder locomotive are at right angles and are 325 mm long. The cylinders are 675 mm apart. The rotating mass per cylinder is 200 kg at the crank pin and the mass of the reciprocating parts per cylinder is 240 kg. The wheel center lines are 1.5 m apart. The whole of the rotating and 2/3rd of the reciprocating masses are to be balanced and the balance masses are to be placed in the plane of the rotation of the driving wheels at a distance of 800 mm. Find: (i) the magnitude and direction of the balancing masses (ii) the magnitude of hammer blow (iii) variation in tractive force and (iv) maximum swaying coupleat a crank speed of 240 rpm.

VIII. An open belt running over two pulleys 24 cm and 60 cm diameters connects two parallel shafts 3 m apart and transmits 3.75 kW from the smaller pulley that rotates at 300 rpm. Coefficient of friction between belt and the pulleys is 0.3 and the safe working tension is 100 N/cm width. Determine (i) minimum width of the belt (ii) initial belt tension and (iii) length of the belt required.

OR

IX. Explain: (i) Rope brake dynamometers (ii) Belt transmission dynamometers.

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